

EN3160: Image Processing and Machine Vision

Assignment 1: Intensity Transformations and Neighborhood Filtering

Department of Electronic and Telecommunication Engineering

University of Moratuwa

September 14, 2026

Name: Samarasinghe S.M.R.R.

Index Number: 230566U

GitHub Repository: github.com/RumethRanhinda/EN3160-Vision-UoM

1. Intensity Transformation Implementation

```
[34]: import cv2
import numpy as np
import matplotlib.pyplot as plt

def intensity_transform(im, breakpoints):
    bp = np.array(breakpoints)
    xp = bp[:, 0] # Input intensities
    yp = bp[:, 1] # Output intensities

    # Apply interpolation and reshape back to original image dimensions
    transformed_im = np.interp(im.flatten(), xp, yp).reshape(im.shape)

    return transformed_im.astype(np.uint8)

img = cv2.imread('Images/emma.jpeg', cv2.IMREAD_GRAYSCALE)

if img is not None:
    # Provided breakpoints
    bp_given = [[0, 0], [49, 49], [50, 100], [149, 255], [150, 150], [255, 255]]

    # Custom breakpoints
    bp_custom = [[0, 0], [50, 20], [175, 100], [255, 255]]

    transformed_given = intensity_transform(img, bp_given)
    transformed_custom = intensity_transform(img, bp_custom)

    # Define grid ticks for the plots
    grid_ticks = [0, 50, 100, 150, 200, 255]
```

```

fig, axes = plt.subplots(2, 3, figsize=(15, 9))

# Given Breakpoints
bp_given_arr = np.array(bp_given)
axes[0, 0].plot(bp_given_arr[:,0], bp_given_arr[:,1], marker='o',
color='#55499C')
axes[0, 0].set_title("Transformation Plot (Given)")
axes[0, 0].set_xlabel("Input intensity"); axes[0, 0].set_ylabel("Output
intensity")
axes[0, 0].set_xlim([0, 255]); axes[0, 0].set_ylim([0, 260])

# Apply custom grid
axes[0, 0].set_xticks(grid_ticks)
axes[0, 0].set_yticks(grid_ticks)
axes[0, 0].grid(True)

axes[0, 1].imshow(img, cmap='gray', vmin=0, vmax=255)
axes[0, 1].set_title("Original Image")
axes[0, 1].axis('off')

axes[0, 2].imshow(transformed_given, cmap='gray', vmin=0, vmax=255)
axes[0, 2].set_title("Transformed Image (Given)")
axes[0, 2].axis('off')

# Custom Breakpoints
bp_custom_arr = np.array(bp_custom)
axes[1, 0].plot(bp_custom_arr[:,0], bp_custom_arr[:,1], marker='o',
color='#55499C')
axes[1, 0].set_title("Transformation Plot (Custom)")
axes[1, 0].set_xlabel("Input intensity"); axes[1, 0].set_ylabel("Output
intensity")
axes[1, 0].set_xlim([0, 255]); axes[1, 0].set_ylim([0, 260])

axes[1, 0].set_xticks(grid_ticks)
axes[1, 0].set_yticks(grid_ticks)
axes[1, 0].grid(True)

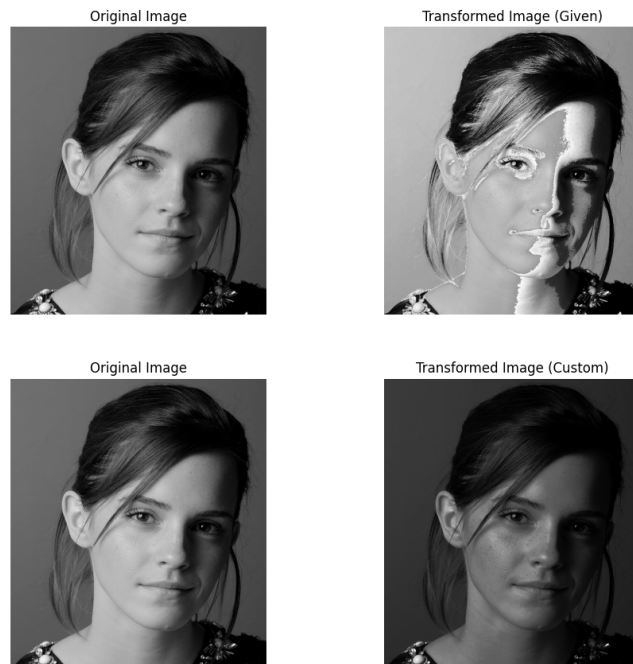
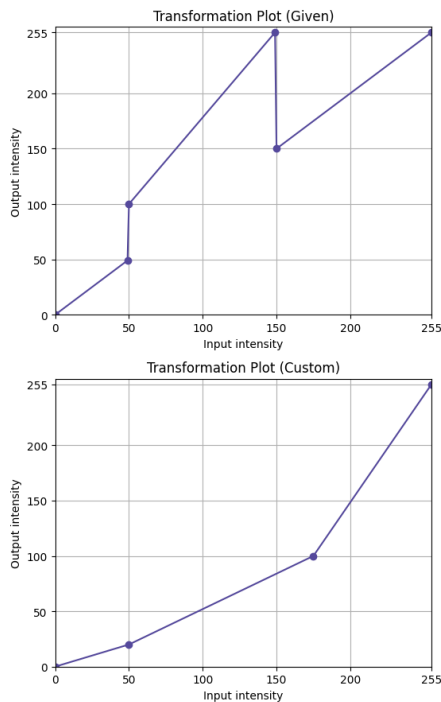
axes[1, 1].imshow(img, cmap='gray', vmin=0, vmax=255)
axes[1, 1].set_title("Original Image")
axes[1, 1].axis('off')

axes[1, 2].imshow(transformed_custom, cmap='gray', vmin=0, vmax=255)
axes[1, 2].set_title("Transformed Image (Custom)")
axes[1, 2].axis('off')

plt.tight_layout()
plt.show()

```

```
else:
    print("Image not found")
```



The given intensity transformation made high intensities darker in a discontinuous way which explains the artifact seen in the output image.

The custom transformation intended to highlight the freckles in the face by providing a wider range for brighter pixels.

2. Accentuate White and Gray Matter

```
[49]: brain_img = cv2.imread('Images/brain.jpeg', cv2.IMREAD_GRAYSCALE)

if brain_img is not None:
    bp_white_matter = [[0, 0], [150, 5], [255, 255]]
    bp_gray_matter = [[0, 0], [128, 0], [190, 255], [255, 255]]
    wm_accentuated = intensity_transform(brain_img, bp_white_matter)
    gm_accentuated = intensity_transform(brain_img, bp_gray_matter)
    grid_ticks = [0, 50, 100, 150, 200, 255]
    fig, axes = plt.subplots(2, 2, figsize=(10, 10))

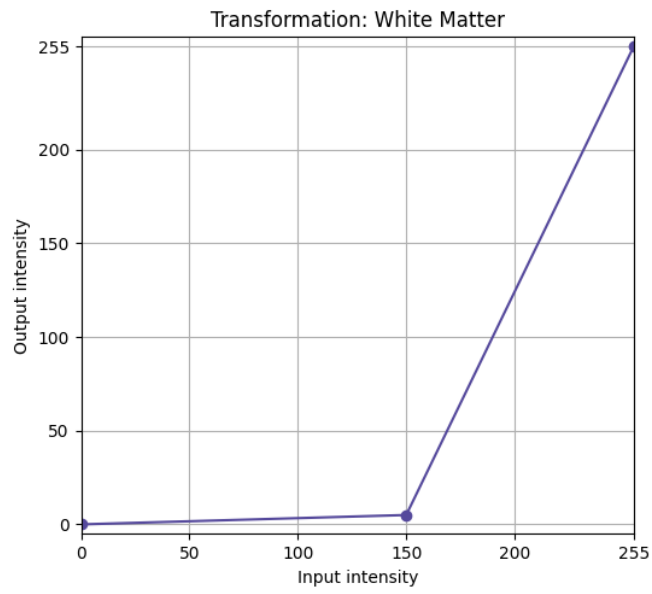
    # White Matter Transformation
    bp_wm_arr = np.array(bp_white_matter)
    axes[0, 0].plot(bp_wm_arr[:,0], bp_wm_arr[:,1], marker='o', color='#55499C')
```

```

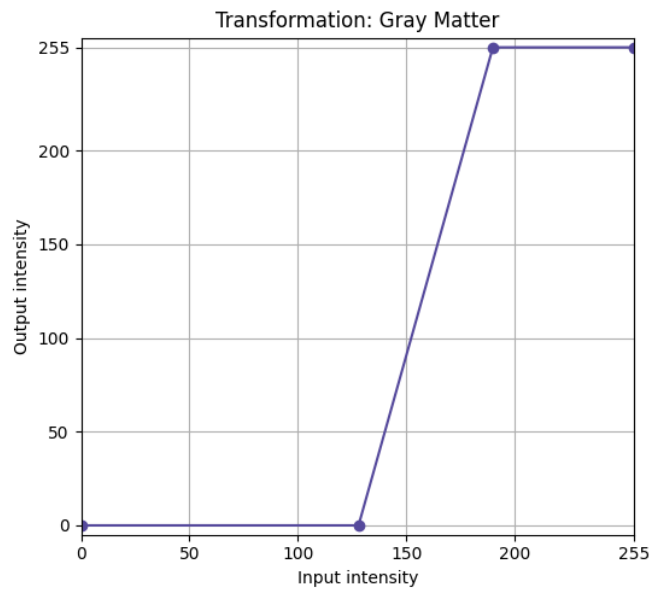
axes[0, 0].set_title("Transformation: White Matter")
axes[0, 0].set_xlabel("Input intensity") ; axes[0, 0].set_ylabel("Output_
↪intensity")
axes[0, 0].set_xlim([0, 255]); axes[0, 0].set_ylim([-5, 260])
axes[0, 0].set_xticks(grid_ticks); axes[0, 0].set_yticks(grid_ticks);
↪axes[0, 0].grid(True)
axes[0, 1].imshow(wm_accentuated, cmap='gray', vmin=0, vmax=255)
axes[0, 1].set_title("White Matter Accentuated"); axes[0, 1].axis('off')

# Gray Matter Transformation
bp_gm_arr = np.array(bp_gray_matter)
axes[1, 0].plot(bp_gm_arr[:,0], bp_gm_arr[:,1], marker='o', color='#55499C')
axes[1, 0].set_title("Transformation: Gray Matter"); axes[1, 0].
↪set_xlabel("Input intensity")
axes[1, 0].set_ylabel("Output intensity"); axes[1, 0].set_xlim([0, 255]);
↪axes[1, 0].set_ylim([-5, 260])
axes[1, 0].set_xticks(grid_ticks); axes[1, 0].set_yticks(grid_ticks);
↪axes[1, 0].grid(True)
axes[1, 1].imshow(gm_accentuated, cmap='gray', vmin=0, vmax=255)
axes[1, 1].set_title("Gray Matter Accentuated"); axes[1, 1].axis('off')
plt.tight_layout()
plt.show()
else:
    print("Image not found")

```



White Matter Accentuated



Gray Matter Accentuated

